Lesson 9: Probability Inequalities

AI/ML Mathematics

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1 Motivation and Overview

How likely is a sample average to deviate from its expectation by more than ε? How fast do empirical estimates of value functions converge? These questions are answered by *probability inequalities*—also called concentration inequalities—which bound tail probabilities in terms of moments, ranges, or structural properties of random variables.

This lesson develops the fundamental chain of inequalities:

1. **Markov's inequality**: the weakest but most general bound (requires only a finite first moment).
2. **Chebyshev's inequality**: uses variance for a tighter bound.
3. **Cauchy–Schwarz inequality**: bounds covariances and expected products.
4. **Hoeffding's inequality**: exponentially tight bound for sums of bounded independent random variables.
5. **Azuma–Hoeffding inequality**: extends Hoeffding to martingale difference sequences.
6. **Jensen's inequality**: relates expectations of convex functions to convex functions of expectations.

Why This Matters for RL

- **Chebyshev** → weak law of large numbers → justification of Monte Carlo estimation.
- **Hoeffding** → PAC (probably approximately correct) sample complexity bounds for multi-armed bandits.
- **Azuma–Hoeffding** → finite-sample bounds for TD error accumulation (the partial sums of TD errors form a martingale, Lesson 8).
- **Jensen** → non-expansiveness of the Bellman operator in weighted norms, and the contraction argument for TD fixed points.

Prerequisites. Lesson 5 (expectation, linearity), Lesson 6 (conditional expectation, tower property), Lesson 7 (variance, independence), and Lesson 8 (filtrations, MDS).

2 Markov's and Chebyshev's Inequalities

2.1 Markov's Inequality

Markov's Inequality

Theorem 2.1 (Markov's Inequality)**.** *If $X \geq 0$ is a non-negative random variable with $\mathbb{E}[X] < \infty$, then for any $a > 0$:*

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}. \tag{1}$$

Proof. Since $X \geq 0$, the inequality $X \geq a \cdot \mathbf{1}_{\{X \geq a\}}$ holds pointwise: when $X \geq a$, the right-hand side equals $a \leq X$; when $X < a$, the right-hand side equals $0 \leq X$. Taking expectations:

$$\mathbb{E}[X] \geq a \cdot \mathbb{E}[\mathbf{1}_{\{X \geq a\}}] = a \cdot \mathbb{P}(X \geq a). \tag{2}$$

Dividing both sides by $a > 0$ gives the result. □

Remark 2.2. Markov's inequality is the weakest concentration inequality but requires the least: only non-negativity and a finite first moment. Every stronger inequality in this lesson is derived from Markov by applying it to a cleverly chosen transformation of the original variable.

Example 2.3. Let X be the outcome of a fair six-sided die. $\mathbb{E}[X] = 3.5$. Markov gives $\mathbb{P}(X \geq 5) \leq 3.5/5 = 0.7$. The true probability is $\mathbb{P}(X \geq 5) = 2/6 \approx 0.333$. The bound is loose but valid.

2.2 Chebyshev's Inequality

Chebyshev's Inequality

Theorem 2.4 (Chebyshev's Inequality)**.** *If X has finite variance $\sigma^2 = \mathrm{Var}(X)$, then for any $\varepsilon > 0$:*

$$\mathbb{P}(|X - \mathbb{E}[X]| \geq \varepsilon) \leq \frac{\mathrm{Var}(X)}{\varepsilon^2}. \tag{3}$$

Proof. Apply Markov's inequality (Theorem 2.1) to the non-negative random variable $(X - \mathbb{E}[X])^2$ with threshold ε^2:

$$\begin{aligned}\mathbb{P}(|X - \mathbb{E}[X]| \geq \varepsilon) &= \mathbb{P}\big((X - \mathbb{E}[X])^2 \geq \varepsilon^2\big) \\ &\leq \frac{\mathbb{E}[(X - \mathbb{E}[X])^2]}{\varepsilon^2} = \frac{\mathrm{Var}(X)}{\varepsilon^2}.\end{aligned}$$

□

Corollary 2.5 (Chebyshev for Sample Mean)**.** *If $X_1, \ldots, X_n$ are i.i.d. with mean μ and variance σ^2, then for $\bar{X}_n = \frac{1}{n}\sum_{i=1}^n X_i$:*

$$\mathbb{P}(|\bar{X}_n - \mu| \geq \varepsilon) \leq \frac{\sigma^2}{n\varepsilon^2}. \tag{4}$$

Proof. $\mathrm{Var}(\bar{X}_n) = \sigma^2/n$ (by Lesson 7, Corollary 2.5). Apply Theorem 2.4 with $X = \bar{X}_n$. □

Remark 2.6. This corollary immediately gives the weak law of large numbers: $\bar{X}_n \xrightarrow{\mathbb{P}} \mu$ as $n \to \infty$, since the right-hand side tends to 0 for any fixed $\varepsilon > 0$. This justifies Monte Carlo estimation of $V^\pi(s)$ from sample returns.

3 Cauchy–Schwarz Inequality for Expectations

Theorem 3.1 (Cauchy–Schwarz Inequality)**.** *For random variables X, Y with $\mathbb{E}[X^2], \mathbb{E}[Y^2] < \infty$:*

$$|\mathbb{E}[XY]| \leq \sqrt{\mathbb{E}[X^2]\,\mathbb{E}[Y^2]}. \tag{5}$$

Equivalently, $(\mathbb{E}[XY])^2 \leq \mathbb{E}[X^2]\,\mathbb{E}[Y^2]$. Equality holds if and only if $Y = cX$ a.s. for some constant $c \in \mathbb{R}$ (or $X = 0$ a.s.).

Proof. If $\mathbb{E}[Y^2] = 0$, then $Y = 0$ a.s., so $\mathbb{E}[XY] = 0$ and the inequality is trivial. Assume $\mathbb{E}[Y^2] > 0$.

For any $t \in \mathbb{R}$, the random variable $(X + tY)^2 \geq 0$, so

$$0 \leq \mathbb{E}[(X + tY)^2] = \mathbb{E}[X^2] + 2t\,\mathbb{E}[XY] + t^2\,\mathbb{E}[Y^2]. \tag{6}$$

This is a quadratic polynomial $q(t) = At^2 + Bt + C$ with $A = \mathbb{E}[Y^2] > 0$, $B = 2\mathbb{E}[XY]$, $C = \mathbb{E}[X^2]$. Since $q(t) \geq 0$ for all t, its discriminant must be non-positive:

$$B^2 - 4AC \leq 0 \quad\Longrightarrow\quad 4(\mathbb{E}[XY])^2 - 4\mathbb{E}[X^2]\mathbb{E}[Y^2] \leq 0 \quad\Longrightarrow\quad (\mathbb{E}[XY])^2 \leq \mathbb{E}[X^2]\mathbb{E}[Y^2]. \tag{7}$$

Taking square roots gives (5).

Equality holds iff the discriminant is zero, iff $q(t) = 0$ for some t^*, iff $(X + t^*Y)^2 = 0$ a.s., iff $X = -t^*Y$ a.s. □

Corollary 3.2 (Covariance Bound). *For random variables X, Y with positive variances:*

$$|\operatorname{Cov}(X,Y)| \le \sqrt{\operatorname{Var}(X)\operatorname{Var}(Y)}. \tag{8}$$

Equivalently, the correlation coefficient satisfies $|\rho(X,Y)| \le 1$. Equality holds iff $Y = aX + b$ a.s. for some $a \neq 0, b \in \mathbb{R}$.

Proof. Apply Theorem 3.1 to the centered variables $\tilde{X} = X - \mathbb{E}[X]$ and $\tilde{Y} = Y - \mathbb{E}[Y]$:

$$|\mathbb{E}[\tilde{X}\tilde{Y}]| \le \sqrt{\mathbb{E}[\tilde{X}^2]\,\mathbb{E}[\tilde{Y}^2]}, \tag{9}$$

which is $|\operatorname{Cov}(X,Y)| \le \sqrt{\operatorname{Var}(X)\operatorname{Var}(Y)}$. □

4 Hoeffding's Lemma and Inequality

4.1 Hoeffding's Lemma

Lemma 4.1 (Hoeffding's Lemma). *If X is a random variable with $\mathbb{E}[X] = 0$ and $a \le X \le b$ a.s., then for all $\lambda > 0$:*

$$\mathbb{E}[e^{\lambda X}] \le \exp\left(\frac{\lambda^2(b-a)^2}{8}\right). \tag{10}$$

Proof. **Step 1: Convexity bound.** Since $a \le X \le b$, we can write X as a convex combination of a and b: $X = \frac{b-X}{b-a} \cdot a + \frac{X-a}{b-a} \cdot b$. By the convexity of $t \mapsto e^{\lambda t}$:

$$e^{\lambda X} \le \frac{b-X}{b-a}\, e^{\lambda a} + \frac{X-a}{b-a}\, e^{\lambda b}. \tag{11}$$

Step 2: Take expectations. Using $\mathbb{E}[X] = 0$:

$$\begin{aligned}\mathbb{E}[e^{\lambda X}] &\le \frac{b-\mathbb{E}[X]}{b-a}\, e^{\lambda a} + \frac{\mathbb{E}[X]-a}{b-a}\, e^{\lambda b} \\ &= \frac{b}{b-a}\, e^{\lambda a} + \frac{-a}{b-a}\, e^{\lambda b}.\end{aligned} \tag{12}$$

Step 3: Auxiliary function. Let $p = -a/(b-a)$ (so $1-p = b/(b-a)$) and $u = \lambda(b-a)$. Then the right-hand side of (12) equals $e^{g(u)}$ where

$$g(u) = -pu + \ln(1 - p + p\,e^u). \tag{13}$$

We verify: $g(0) = \ln 1 = 0$ and

$$g'(u) = -p + \frac{p\,e^u}{1-p+p\,e^u}, \qquad g'(0) = -p + p = 0. \tag{14}$$

For the second derivative:

$$g''(u) = \frac{p(1-p)\,e^u}{(1-p+p\,e^u)^2}. \tag{15}$$

Setting $q = pe^u/(1-p+pe^u) \in (0,1)$, we have $g''(u) = q(1-q) \le 1/4$ (since $q(1-q) \le 1/4$ for all $q \in [0,1]$ by AM-GM).

Step 4: Taylor bound. By Taylor's theorem with Lagrange remainder, there exists $\xi \in (0,u)$ such that $g(u) = g(0) + g'(0)u + \frac{1}{2}g''(\xi)u^2 = \frac{1}{2}g''(\xi)u^2 \le u^2/8$. Substituting $u = \lambda(b-a)$:

$$\mathbb{E}[e^{\lambda X}] \le e^{g(u)} \le e^{u^2/8} = \exp\left(\frac{\lambda^2(b-a)^2}{8}\right). \tag{16}$$

□

4.2 Hoeffding's Inequality

Hoeffding's Inequality

Theorem 4.2 (Hoeffding's Inequality). *Let $X_1, \ldots, X_n$ be independent random variables with $a_i \le X_i \le b_i$ a.s. Let $\bar{X} = \frac{1}{n}\sum_{i=1}^n X_i$. Then for any $t > 0$:*

$$\mathbb{P}\big(|\bar{X} - \mathbb{E}[\bar{X}]| \ge t\big) \le 2\exp\left(-\frac{2n^2t^2}{\sum_{i=1}^n (b_i - a_i)^2}\right). \tag{17}$$

Proof. Let $Y_i = X_i - \mathbb{E}[X_i]$ (so $\mathbb{E}[Y_i] = 0$ and $Y_i \in [a_i - \mathbb{E}[X_i], b_i - \mathbb{E}[X_i]]$, an interval of length $b_i - a_i$). Let $S = \sum_{i=1}^n Y_i$. We bound $\mathbb{P}(S \ge nt)$; the bound for $\mathbb{P}(S \le -nt)$ follows symmetrically.

Step 1: Chernoff bound (Markov for exponentials). For any $\lambda > 0$:

$$\mathbb{P}(S \ge nt) = \mathbb{P}(e^{\lambda S} \ge e^{\lambda nt}) \le e^{-\lambda nt}\,\mathbb{E}[e^{\lambda S}]. \tag{18}$$

Step 2: Factor the moment generating function. Since the Y_i are independent:

$$\mathbb{E}[e^{\lambda S}] = \mathbb{E}\left[e^{\lambda \sum_i Y_i}\right] = \prod_{i=1}^n \mathbb{E}[e^{\lambda Y_i}]. \tag{19}$$

Step 3: Apply Hoeffding's Lemma. Each Y_i has mean 0 and range $b_i - a_i$. By Lemma 4.1:

$$\mathbb{E}[e^{\lambda Y_i}] \le \exp\left(\frac{\lambda^2(b_i - a_i)^2}{8}\right). \tag{20}$$

Therefore:

$$\mathbb{E}[e^{\lambda S}] \le \exp\left(\frac{\lambda^2}{8}\sum_{i=1}^n (b_i - a_i)^2\right). \tag{21}$$

Step 4: Optimize over λ. Combining (18) and (21):

$$\mathbb{P}(S \ge nt) \le \exp\left(-\lambda nt + \frac{\lambda^2}{8}\sum_i (b_i - a_i)^2\right). \tag{22}$$

The exponent $h(\lambda) = -\lambda nt + \frac{\lambda^2}{8}\sum_i (b_i - a_i)^2$ is a quadratic in λ with minimum at $\lambda^* = 4nt/\sum_i (b_i - a_i)^2$. Substituting:

$$h(\lambda^*) = -\frac{2n^2t^2}{\sum_i (b_i - a_i)^2}. \tag{23}$$

Thus $\mathbb{P}(S \ge nt) \le \exp\big(-2n^2t^2/\sum_i (b_i - a_i)^2\big)$.

Step 5: Two-sided bound. Applying the same argument to $-S$ (or equivalently to $-Y_i$, which has the same range): $\mathbb{P}(S \le -nt) \le \exp(-2n^2t^2/\sum_i (b_i - a_i)^2)$. By the union bound:

$$\mathbb{P}(|S| \ge nt) \le 2\exp\left(-\frac{2n^2t^2}{\sum_i (b_i - a_i)^2}\right). \tag{24}$$

□

Remark 4.3 (Special case: identical ranges). If all $X_i \in [a, b]$ (same range), then $\sum_i (b_i - a_i)^2 = n(b - a)^2$, and Hoeffding's inequality simplifies to

$$\mathbb{P}(|\bar{X} - \mathbb{E}[\bar{X}]| \ge t) \le 2\exp\left(-\frac{2nt^2}{(b - a)^2}\right). \tag{25}$$

This is an exponentially decreasing bound in n, far tighter than Chebyshev's $O(1/n)$ bound.

Connection to RL: Sample Complexity

In a multi-armed bandit with K arms, each arm's reward lies in $[0,1]$. After n pulls of arm a, the empirical mean reward $\hat{\mu}_a$ satisfies $\mathbb{P}(|\hat{\mu}_a - \mu_a| \geq \varepsilon) \leq 2e^{-2n\varepsilon^2}$ by Hoeffding. Setting the right-hand side equal to δ and solving: $n \geq \frac{1}{2\varepsilon^2} \ln \frac{2}{\delta}$ samples suffice to estimate μ_a to within ε with probability at least $1-\delta$. This is the foundation of PAC sample complexity bounds.

5 Azuma–Hoeffding Inequality for Martingale Differences

Azuma–Hoeffding Inequality

Theorem 5.1 (Azuma–Hoeffding Inequality). *Let $\{M_k\}_{k=0}^n$ be a martingale with respect to $\{\mathcal{F}_k\}$ such that $|M_k - M_{k-1}| \leq c_k$ a.s. for deterministic constants $c_k > 0$, $k = 1, \ldots, n$. Then for any $t > 0$:*

$$\mathbb{P}(|M_n - M_0| \geq t) \leq 2\exp\left(-\frac{t^2}{2\sum_{k=1}^n c_k^2}\right). \tag{26}$$

Proof. Define the martingale differences $D_k = M_k - M_{k-1}$ for $k = 1, \ldots, n$. Then $\mathbb{E}[D_k \mid \mathcal{F}_{k-1}] = 0$ and $|D_k| \leq c_k$ a.s., so $D_k \in [-c_k, c_k]$.

Step 1: Chernoff bound. For any $\lambda > 0$:

$$\mathbb{P}(M_n - M_0 \geq t) = \mathbb{P}\left(e^{\lambda(M_n - M_0)} \geq e^{\lambda t}\right) \leq e^{-\lambda t}\,\mathbb{E}[e^{\lambda(M_n - M_0)}]. \tag{27}$$

Step 2: Iterated conditioning. Write $M_n - M_0 = \sum_{k=1}^n D_k$. We bound the MGF by peeling off the outermost term:

$$\mathbb{E}\left[e^{\lambda \sum_{k=1}^n D_k}\right] = \mathbb{E}\left[e^{\lambda \sum_{k=1}^{n-1} D_k} \cdot \mathbb{E}\left[e^{\lambda D_n} \,\middle|\, \mathcal{F}_{n-1}\right]\right]. \tag{28}$$

This uses the tower property: take the outer expectation of $\mathbb{E}[\cdot \mid \mathcal{F}_{n-1}]$, and pull out the $\mathcal{F}_{n-1}$-measurable factor $e^{\lambda \sum_{k=1}^{n-1} D_k}$.

Step 3: Conditional Hoeffding's Lemma. Conditionally on $\mathcal{F}_{n-1}$, D_n has conditional mean 0 and $D_n \in [-c_n, c_n]$ (an interval of length $2c_n$). Hoeffding's Lemma (Lemma 4.1) applied conditionally gives:

$$\mathbb{E}[e^{\lambda D_n} \mid \mathcal{F}_{n-1}] \leq \exp\left(\frac{\lambda^2 (2c_n)^2}{8}\right) = \exp\left(\frac{\lambda^2 c_n^2}{2}\right). \tag{29}$$

Step 4: Iterate. Substituting (29) into (28):

$$\mathbb{E}\left[e^{\lambda \sum_{k=1}^n D_k}\right] \leq e^{\lambda^2 c_n^2/2}\,\mathbb{E}\left[e^{\lambda \sum_{k=1}^{n-1} D_k}\right]. \tag{30}$$

Repeating for $k = n-1, n-2, \ldots, 1$:

$$\mathbb{E}\left[e^{\lambda(M_n - M_0)}\right] \leq \exp\left(\frac{\lambda^2}{2}\sum_{k=1}^n c_k^2\right). \tag{31}$$

Step 5: Optimize. Combining (27) and (31): $\mathbb{P}(M_n - M_0 \geq t) \leq \exp(-\lambda t + \lambda^2 \sum c_k^2/2)$. Minimizing over λ: $\lambda^* = t/\sum c_k^2$, giving

$$\mathbb{P}(M_n - M_0 \geq t) \leq \exp\left(-\frac{t^2}{2\sum_{k=1}^n c_k^2}\right). \tag{32}$$

Step 6: Two-sided bound. Since $-M_k$ is also a martingale with $|(-M_k) - (-M_{k-1})| = |D_k| \le c_k$, the same bound applies to $\mathbb{P}(M_0 - M_n \ge t)$. By the union bound:

$$\mathbb{P}(|M_n - M_0| \ge t) \le 2\exp\left(-\frac{t^2}{2\sum_{k=1}^n c_k^2}\right). \tag{33}$$

□

Remark 5.2 (Comparison with Hoeffding). If the martingale differences D_k are independent (i.e., the martingale is a random walk), Azuma–Hoeffding recovers Hoeffding's inequality. The power of Azuma–Hoeffding is that it does *not* require independence—only the MDS property $\mathbb{E}[D_k \mid \mathcal{F}_{k-1}] = 0$ and bounded increments.

Connection to RL: TD Error Concentration

In a finite MDP with $|R_{t+1}| \le R_{\max}$ and $\gamma \in [0,1)$, the TD error $\delta_t^\pi = R_{t+1} + \gamma V^\pi(S_{t+1}) - V^\pi(S_t)$ satisfies $|\delta_t^\pi| \le 2R_{\max}/(1-\gamma)$. The partial sums $M_n = \sum_{t=0}^{n-1} \delta_t^\pi$ form a martingale (Lesson 8), and Azuma–Hoeffding gives:

$$\mathbb{P}\left(\left|\sum_{t=0}^{n-1} \delta_t^\pi\right| \ge \varepsilon\right) \le 2\exp\left(-\frac{\varepsilon^2(1-\gamma)^2}{8nR_{\max}^2}\right).$$

This provides finite-sample concentration bounds for accumulated TD errors, complementing the asymptotic convergence guaranteed by stochastic approximation theory.

6 Jensen's Inequality

6.1 Convex Functions

Definition 6.1 (Convex Function)**.** A function $\varphi : I \to \mathbb{R}$ (where $I \subseteq \mathbb{R}$ is an interval) is *convex* if for all $x, y \in I$ and $\lambda \in [0,1]$:

$$\varphi(\lambda x + (1-\lambda)y) \le \lambda\,\varphi(x) + (1-\lambda)\,\varphi(y). \tag{34}$$

The function is *strictly convex* if strict inequality holds whenever $x \ne y$ and $\lambda \in (0,1)$.

Example 6.2. Common convex functions: x^2, $|x|$, e^x, $-\ln x$ (on $x > 0$), $x \ln x$ (on $x > 0$). Common concave functions: $\ln x$, $\sqrt{x}$ (on $x \ge 0$).

6.2 Supporting Hyperplane Lemma

Lemma 6.3 (Supporting Hyperplane)**.** *If $\varphi : I \to \mathbb{R}$ is convex, then for any $c \in I$, there exists $m \in \mathbb{R}$ (a subgradient at c) such that*

$$\varphi(x) \ge \varphi(c) + m(x-c) \quad \textit{for all } x \in I. \tag{35}$$

Proof. Define the left-derivative slope:

$$m = \sup_{y<c} \frac{\varphi(c) - \varphi(y)}{c - y}. \tag{36}$$

By convexity, the slopes of secant lines through $(c, \varphi(c))$ are non-decreasing: if $y_1 < y_2 < c$, then

$$\frac{\varphi(c) - \varphi(y_1)}{c - y_1} \le \frac{\varphi(c) - \varphi(y_2)}{c - y_2}. \tag{37}$$

Similarly, for $x > c$, the secant slope from c to x is at least m:

$$\frac{\varphi(x) - \varphi(c)}{x - c} \geq m, \tag{38}$$

which gives $\varphi(x) \geq \varphi(c) + m(x - c)$.

For $x < c$, the secant slope from x to c satisfies:

$$\frac{\varphi(c) - \varphi(x)}{c - x} \leq m, \tag{39}$$

which rearranges to $\varphi(x) \geq \varphi(c) + m(x - c)$. The case $x = c$ is trivial. □

6.3 Jensen's Inequality

Jensen's Inequality

Theorem 6.4 (Jensen's Inequality)**.** *Let φ be convex and let X be a random variable with $\mathbb{E}[|X|] < \infty$ and $\mathbb{E}[|\varphi(X)|] < \infty$. Then:*

$$\varphi(\mathbb{E}[X]) \leq \mathbb{E}[\varphi(X)]. \tag{40}$$

If φ is concave, the inequality is reversed: $\varphi(\mathbb{E}[X]) \geq \mathbb{E}[\varphi(X)]$.

Proof. Let $c = \mathbb{E}[X]$. By Lemma 6.3, there exists m such that $\varphi(x) \geq \varphi(c) + m(x - c)$ for all x. Applying this pointwise to X:

$$\varphi(X) \geq \varphi(c) + m(X - c). \tag{41}$$

Taking expectations and using linearity:

$$\begin{aligned} \mathbb{E}[\varphi(X)] &\geq \mathbb{E}[\varphi(c) + m(X - c)] \\ &= \varphi(c) + m(\mathbb{E}[X] - c) = \varphi(c) + m \cdot 0 = \varphi(c) = \varphi(\mathbb{E}[X]). \end{aligned} \tag{42}$$

For concave φ, apply the convex case to $-\varphi$. □

Theorem 6.5 (Conditional Jensen's Inequality)**.** *If φ is convex and $\mathbb{E}[|\varphi(X)|] < \infty$, then*

$$\varphi(\mathbb{E}[X \mid Y]) \leq \mathbb{E}[\varphi(X) \mid Y] \quad a.s. \tag{43}$$

Proof. Fix $Y = y$. Apply the unconditional Jensen's inequality to the conditional distribution of X given $Y = y$: $\varphi(\mathbb{E}[X \mid Y = y]) \leq \mathbb{E}[\varphi(X) \mid Y = y]$. Since this holds for every y, it holds as a random variable inequality. □

Connection to RL: Bellman Operator Contraction

Jensen's inequality is used to prove that the Bellman operator $\mathcal{T}^\pi$ is non-expansive in weighted norms. The key step: the transition operator $(\mathbf{P}_\pi v)(s') = \sum_s w_{s's}\, v(s)$ computes a weighted average of $v(s)$ with non-negative weights summing to 1. Since $\varphi(x) = x^2$ is convex, Jensen gives:

$$\left(\sum_s w_{s's}\, v(s)\right)^2 \leq \sum_s w_{s's}\, v(s)^2.$$

Summing over s' with appropriate weights proves $\|\mathbf{P}_\pi v\|_{d^\pi} \leq \|v\|_{d^\pi}$, i.e., $\mathbf{P}_\pi$ is a contraction in the d^π-weighted L^2-norm. This is essential for the convergence of linear TD.

7 Exercises

Exercise 7.1. Let X be a random variable (not necessarily non-negative) with $\mathbb{E}[X^2] < \infty$. Use Markov's inequality applied to X^2 to show that $\mathbb{P}(|X| \geq a) \leq \mathbb{E}[X^2]/a^2$ for any $a > 0$.

Exercise 7.2. Let $X_1, \ldots, X_n$ be i.i.d. Bernoulli(1/2) and $\bar{X}_n = \frac{1}{n}\sum X_i$. Use Chebyshev's inequality to find the smallest n such that $\mathbb{P}(|\bar{X}_n - 1/2| \geq 0.1) \leq 0.05$. Compare with the Hoeffding bound.

Exercise 7.3. Use the Cauchy–Schwarz inequality to prove that if $\mathbb{E}[X^2] < \infty$, then $\mathbb{E}[|X|] < \infty$. *Hint:* Apply Cauchy–Schwarz with $Y = 1$.

Exercise 7.4. A bandit arm has unknown mean $\mu \in [0, 1]$. After n independent pulls with rewards $X_1, \ldots, X_n \in [0, 1]$, we estimate $\hat{\mu} = \bar{X}_n$. Use Hoeffding's inequality to construct a 95% confidence interval for μ as a function of n.

Exercise 7.5. Let $X_1, \ldots, X_n$ be i.i.d. with $\mathbb{P}(X_i = 1) = \mathbb{P}(X_i = -1) = 1/2$. Use Azuma–Hoeffding to bound $\mathbb{P}(|\sum_{i=1}^n X_i| \geq c\sqrt{n})$ for $c > 0$. Compare with the exact probability for $n = 100$, $c = 2$.

Exercise 7.6. Let $X > 0$ a.s. with $\mathbb{E}[X] < \infty$. Use Jensen's inequality with $\varphi(x) = -\ln x$ (convex on $(0, \infty)$) to prove that $\mathbb{E}[\ln X] \leq \ln \mathbb{E}[X]$ (the AM-GM inequality for expectations).

Exercise 7.7. Use Jensen's inequality with $\varphi(x) = x^2$ to give an alternative proof that $\mathrm{Var}(X) \geq 0$, i.e., $\mathbb{E}[X^2] \geq (\mathbb{E}[X])^2$.

Exercise 7.8. Prove the *conditional Markov inequality*: if $X \geq 0$ and $\mathbb{E}[X \mid Y] < \infty$ a.s., then for any $a > 0$:

$$\mathbb{P}(X \geq a \mid Y) \leq \frac{\mathbb{E}[X \mid Y]}{a} \quad \text{a.s.}$$

Hint: Apply the standard proof of Markov's inequality conditionally on Y.

8 Summary

Inequality	**Statement**	**RL Application**
Markov	$\mathbb{P}(X \geq a) \leq \mathbb{E}[X]/a$	Building block for all others
Chebyshev	$\mathbb{P}(\lvert X - \mu\rvert \geq \varepsilon) \leq \sigma^2/\varepsilon^2$	WLLN $\to$ MC justification
Cauchy–Schwarz	$\lvert\mathbb{E}[XY]\rvert \leq \sqrt{\mathbb{E}[X^2]\mathbb{E}[Y^2]}$	Covariance bounds
Hoeffding	$\mathbb{P}(\lvert\bar{X} - \mu\rvert \geq t) \leq 2e^{-2nt^2/(b-a)^2}$	PAC bandit bounds
Azuma–Hoeffding	$\mathbb{P}(\lvert M_n - M_0\rvert \geq t) \leq 2e^{-t^2/(2\sum c_k^2)}$	TD error concentration
Jensen	$\varphi(\mathbb{E}[X]) \leq \mathbb{E}[\varphi(X)]$ (convex)	Bellman contraction

Next lesson: Convergence of Random Variables and the Borel–Cantelli Lemma — the four modes of convergence (a.s., in probability, in L^p, in distribution), their relationships, and the lemma that connects summable probabilities to almost sure events.

References